Robert Paine



Born in 1933, in Cambridge, Massachusetts, Robert Paine studied at Harvard. After a stint in the US Army, where he was the battalion gardener, Paine focused his research on marine invertebrates. His study of the relationship between starfish and mussels on the Pacific coast led him to propose the concept of keystone species—the disproportionate impact that a single species can have on its ecosystem.

Paine worked for most of his career at the University of Washington, where he popularized field manipulation experiments, or "kick-it-and-see" ecology. He was awarded the International Cosmos Award by the National Academy of Sciences in 2013, and died in 2016.

Key works

1966 "Food Web Complexity and Species Diversity," *American Naturalist*

1969 "A Note on Trophic Complexity and Community Stability," *American Naturalist*

1994 *Marine Rocky Shores and Community Ecology: An Experimentalist's Perspective*